

CSE 331

AUTOMATA AND COMPUTABILITY

Equivalence of DFAs and NFAs

Summer 2026

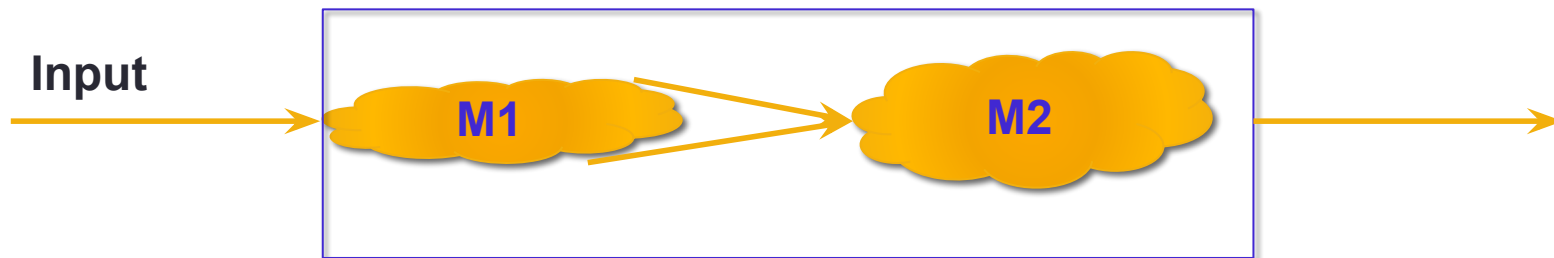
Today's learning goals

Sipser Ch 1.2, 1.3

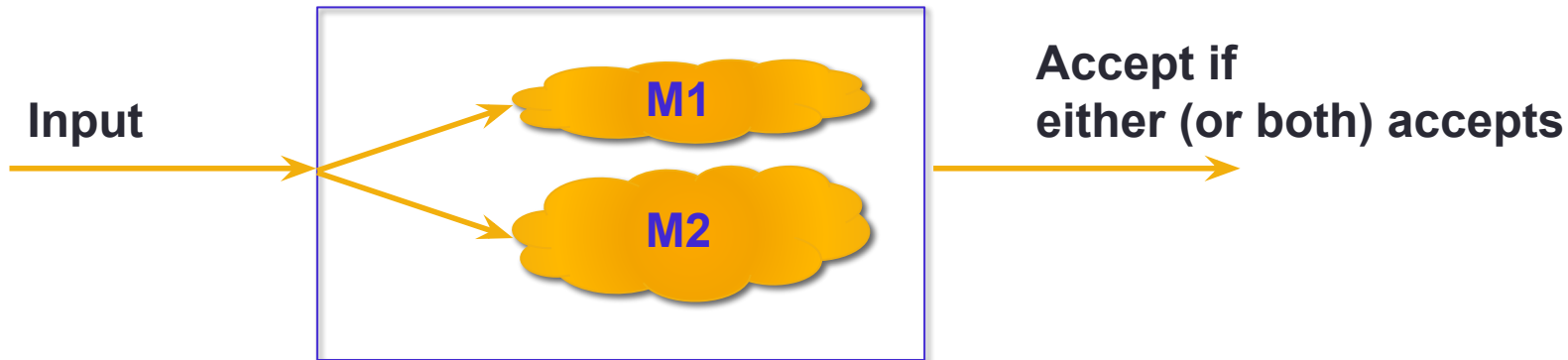
- Design NFA recognizing a given language
- Convert an NFA (with or without spontaneous moves) to a DFA recognizing the same language
- Decide whether or not a string is described by a given regular expression
- Design a regular expression to describe a given language

Nondeterministic finite automata

- "Guess" some stage of input at which switch modes



- "Guess" one of finite list of criteria to meet

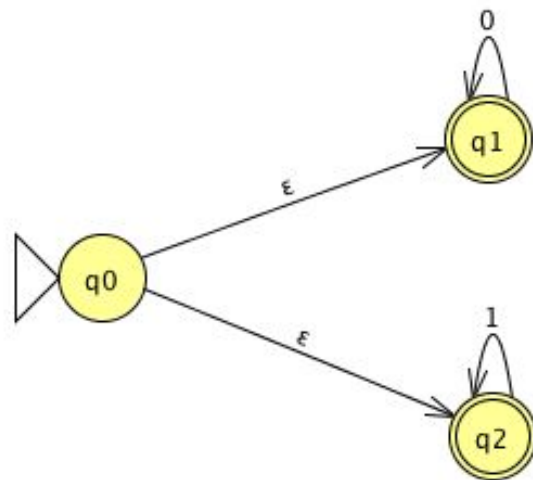
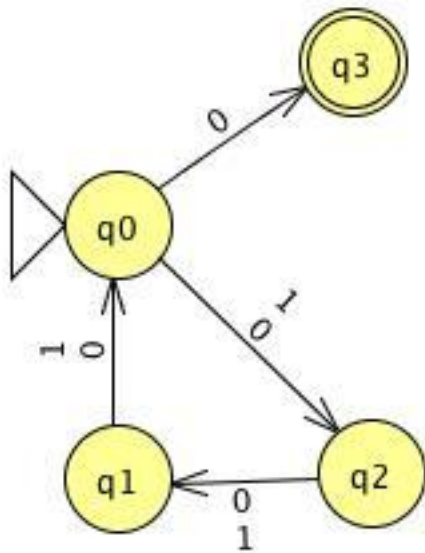


Acceptance in an NFA

An NFA $(Q, \Sigma, \delta, q_0, F)$ accepts a string w in Σ^* iff we can write $w = y_1 y_2 \cdots y_m$ where each $y_i \in \Sigma_\epsilon$ and **there is** a sequence of states $r_0, \dots, r_m \in Q$ such that

1. $r_0 = q_0$
2. $r_{i+1} \in \delta(r_i, y_{i+1})$ for each $i = 0, \dots, m - 1$
3. $r_m \in F$.

NFA



Simulating NFA with DFA

Not quite a closure proof, but ...

Proof:

Given name variables for sets, machines assumed to exist.

WTS state goal and outline plan.

Construction using objects previously defined + new tools working towards goal. Give formal definition and explain.

Correctness prove that construction works.

Conclusion recap what you've proved.

Simulating NFA with DFA

For any language recognized by an NFA, there is a DFA that recognizes this language.

Proof:

Given A, a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

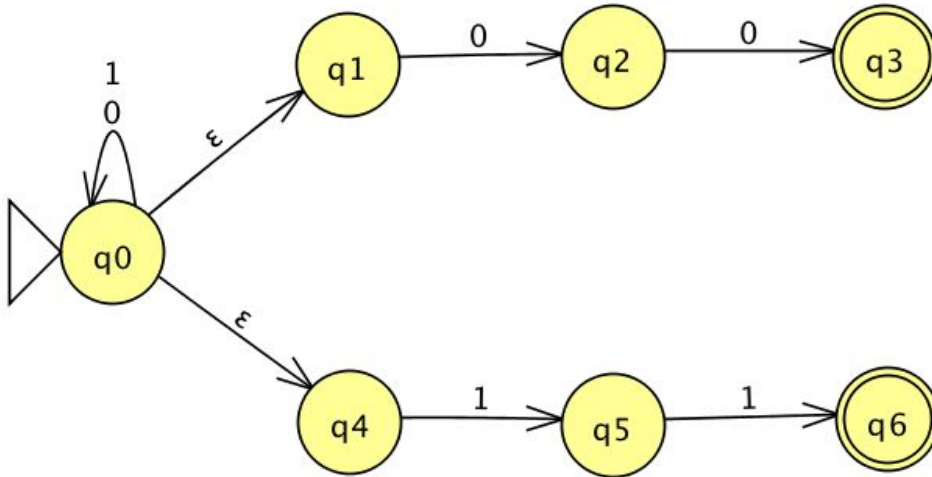
WTS there exists a DFA M with $L(M) = A$

Construction

Correctness

Conclusion

From NFA to DFA



Which of the following strings is accepted by this NFA?

- A. The empty string
- B. 01
- C. 111
- D. 0
- E. More than one of the above.

Subset construction

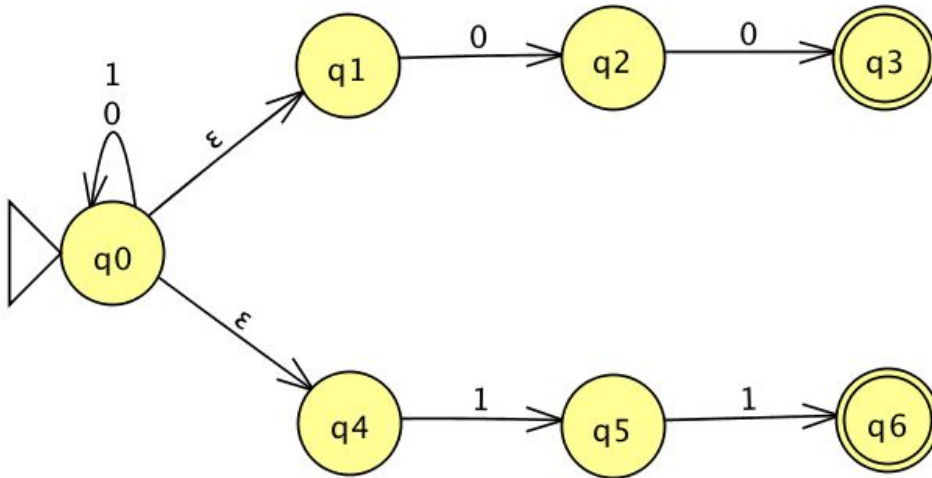
Given A , a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

WTS there exists a DFA M with $L(M) = A$

Construction Define $M = (Q', \Sigma, \delta', q_0', F')$ with

- $Q' = \text{the power set of } Q = \{ X \mid X \text{ is a subset of } Q \}$
- $q_0' = \{ \text{states } N \text{ can be in before first input symbol read} \}$
- $F' = \{ \quad \}$
- $\delta' (\quad) =$

From NFA to DFA



Which states can this NFA be in before first input symbol is read?

- A. q_0
- B. any state
- C. q_0, q_1
- D. q_0, q_4
- E. q_0, q_1, q_4

Subset construction

Given A, a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

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Construction Define $M = (Q', \Sigma, \delta', q_0', F')$ with

- $Q' = \text{the power set of } Q = \{ X \mid X \text{ is a subset of } Q \}$
- $q_0' = \{ q_0 \} \cup \delta(q_0, \epsilon) \cup \dots = \delta^*(q_0, \epsilon)$
- $F' = \{ \quad \}$
- $\delta' (\quad) =$

Subset construction

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Subset construction

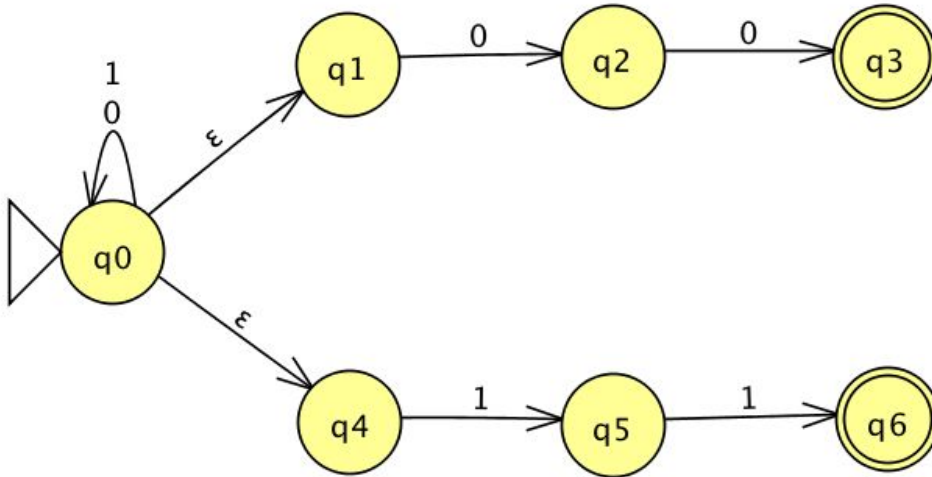
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- $Q' = \text{the power set of } Q = \{ X \mid X \text{ is a subset of } Q \}$
- $q_0' = \delta^*(q_0, \epsilon)$
- $F' = \{ \text{guarantee at least one computation is successful} \}$
- $\delta' (\quad) =$

From NFA to DFA



What does it mean for a set of states X to guarantee at least one computation is successful?

- A. X is a subset of F
- B. $X = F$
- C. $X \cap F$ is nonempty
- D. X is an element of F
- E. None of the above.

Subset construction

Given A , a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

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Construction Define $M = (Q', \Sigma, \delta', q_0', F')$ with

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- $q_0' = \delta^*(q_0, \epsilon)$
- $F' = \{ X \mid X \text{ is a subset of } Q \text{ and } X \cap F \text{ is nonempty} \}$
- $\delta' ((X, x)) = \{ \quad \quad \}$

Subset construction

Given A , a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

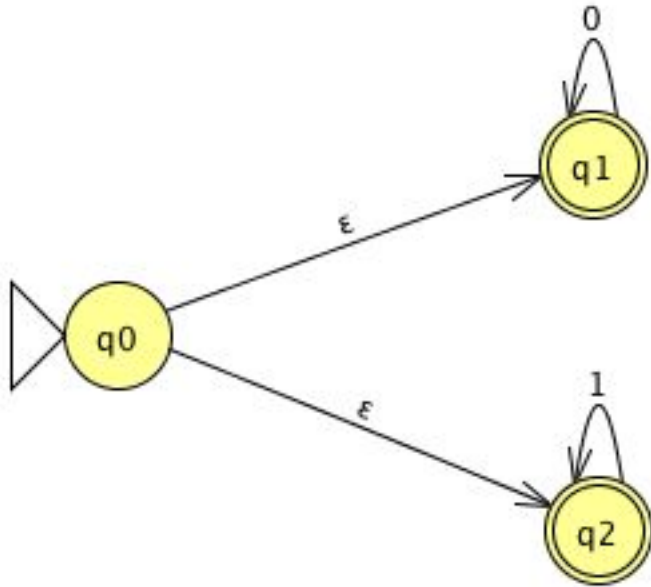
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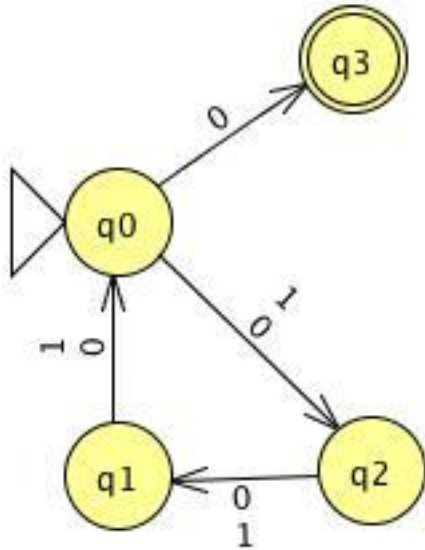
- $Q' =$ the power set of $Q = \{ X \mid X \text{ is a subset of } Q \}$
- $q_0' = \delta^*(q_0, \epsilon)$
- $F' = \{ X \mid X \text{ is a subset of } Q \text{ and } X \cap F \text{ is nonempty} \}$
- $\delta' ((X, x)) = \{ q \text{ in } Q \mid q \text{ is in } \delta(r, x) \text{ for some } r \text{ in } X \text{ or accessible via spontaneous moves} \}$

Types?

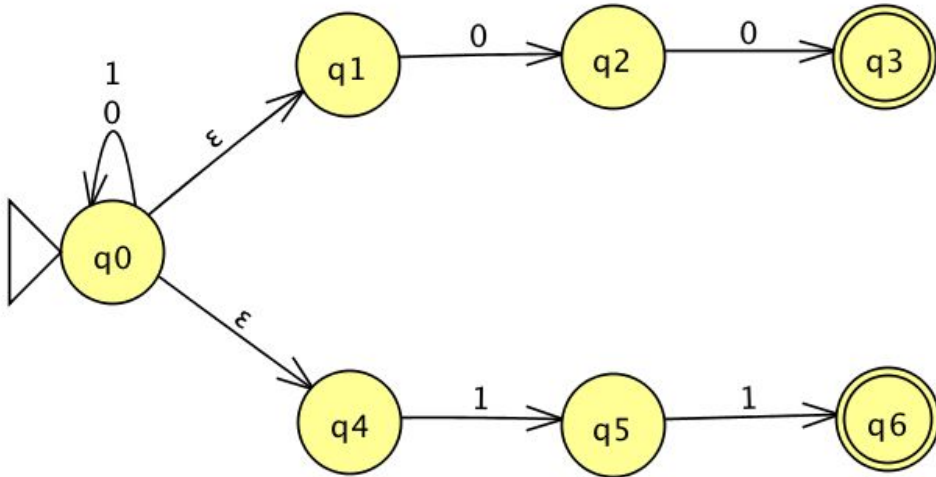
Subset construction examples



Subset construction examples



Subset construction examples



Recap

Regular sets of strings i.e., those that aren't too complicated are those that are

- the language of some DFA, or equivalently,
- the language of some NFA.

If a set of strings can be expressed as the result of **complement, union, intersection, concatenation, Kleene star** of regular languages, then it itself is regular.